

Norway

SLEIPNER

Volve F

F-15

F-15D

Design: F-15D

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15D	Survey Calculation Method:	Minimum Curvature
Design:	F-15D	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-15					
Well Position	+N/-S	-3.17 m	Northing:	6,478,560.36 m	Latitude:	58° 26' 29.7063 N
	+E/-W	3.53 m	Easting:	435,053.55 m	Longitude:	1° 53' 15.1494 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-15D				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	25.06.2013	-1.17	71.62	50,516

Design	F-15D				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	1,364.50
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.17	3.53	212.80	

Survey Program	Date	19.11.2015			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
153.40	1,364.50	12 1/4" F-15 Drop Gyro (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD	
1,389.40	2,599.90	15/9-F-15 D 17 1/2" AutoTrak G3 (F-15D)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
2,630.60	3,275.10	15/9-F-15 D 12 1/4" AutoTrak G3 (F-15D)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
3,354.30	4,671.00	15/9-F-15 D 8 1/2" AutoTrak G3 (F-15D)	Magn, IFR, non-mag, dual in	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.17	3.53	0.00	0.000	0.000	0.000	
153.40	0.15	253.69	153.40	-3.17	3.52	0.01	0.600	0.600	0.000	
234.20	0.28	146.02	234.20	-3.37	3.53	0.17	0.132	0.048	-39.976	
274.60	0.78	136.21	274.60	-3.65	3.78	0.27	0.376	0.371	-7.285	
315.00	1.46	128.29	314.99	-4.16	4.37	0.38	0.517	0.505	-5.881	
355.30	3.41	124.07	355.25	-5.15	5.77	0.46	1.457	1.452	-3.141	
395.70	5.13	126.98	395.54	-6.91	8.21	0.61	1.287	1.277	2.161	
436.10	5.52	139.07	435.76	-9.47	10.92	1.29	0.880	0.290	8.978	

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Wellbore:	F-15D	Survey Calculation Method:	Minimum Curvature
Design:	F-15D	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
474.40	5.62	152.33	473.89	-12.52	13.00	2.73	1.009	0.078	10.386	
514.80	6.19	163.05	514.07	-16.36	14.55	5.11	0.920	0.423	7.960	
557.10	7.03	171.79	556.09	-21.10	15.59	8.54	0.928	0.596	6.199	
597.50	7.74	174.05	596.16	-26.25	16.22	12.53	0.570	0.527	1.678	
637.80	8.49	174.54	636.05	-31.91	16.79	16.98	0.561	0.558	0.365	
678.20	8.94	175.57	675.99	-38.01	17.31	21.82	0.354	0.334	0.765	
718.50	9.34	177.45	715.77	-44.40	17.70	26.98	0.372	0.298	1.400	
758.90	10.22	185.51	755.59	-51.24	17.50	32.84	1.207	0.653	5.985	
799.30	11.89	193.59	795.24	-58.86	16.18	39.96	1.689	1.240	6.000	
839.70	13.68	199.13	834.64	-67.42	13.64	48.53	1.610	1.329	4.114	
880.00	15.14	201.54	873.67	-76.81	10.14	58.32	1.175	1.087	1.794	
920.40	13.72	203.30	912.79	-86.12	6.31	68.22	1.103	-1.054	1.307	
960.80	14.53	201.31	951.97	-95.24	2.57	77.91	0.701	0.601	-1.478	
1,001.10	16.51	200.62	990.80	-105.31	-1.28	88.47	1.480	1.474	-0.514	
1,041.60	16.33	201.19	1,029.65	-116.01	-5.37	99.67	0.179	-0.133	0.422	
1,081.90	16.18	205.25	1,068.34	-126.37	-9.81	110.78	0.853	-0.112	3.022	
1,122.30	17.54	211.88	1,107.01	-136.63	-15.43	122.45	1.747	1.010	4.923	
1,162.60	17.66	221.21	1,145.43	-146.39	-22.66	134.57	2.100	0.089	6.945	
1,203.00	16.05	230.28	1,184.10	-154.57	-31.00	145.97	2.286	-1.196	6.735	
1,243.30	15.33	232.77	1,222.90	-161.35	-39.52	156.29	0.734	-0.536	1.854	
1,283.80	18.17	237.03	1,261.68	-168.03	-49.09	167.08	2.291	2.104	3.156	
1,324.10	21.21	241.08	1,299.62	-174.98	-60.74	179.23	2.479	2.263	3.015	
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	192.97	2.659	2.599	1.448	
1,389.40	21.90	245.03	1,359.68	-186.67	-83.52	201.40	3.517	-3.386	2.410	
1,409.80	21.44	249.87	1,378.64	-189.56	-90.47	207.59	2.713	-0.676	7.118	
1,469.50	19.80	266.07	1,434.56	-194.01	-110.82	222.36	2.973	-0.824	8.142	
1,510.00	18.51	273.32	1,472.82	-194.10	-124.08	229.62	2.002	-0.956	5.369	
1,551.00	16.25	280.28	1,511.95	-192.70	-136.23	235.03	2.244	-1.654	5.093	
1,591.00	15.05	294.29	1,550.48	-189.57	-146.47	237.94	2.966	-0.900	10.507	
1,631.10	15.19	309.35	1,589.21	-184.09	-155.28	238.11	2.933	0.105	11.267	
1,672.10	15.54	316.07	1,628.74	-176.73	-163.25	236.24	1.327	0.256	4.917	
1,712.70	16.89	323.84	1,667.73	-168.05	-170.50	232.87	1.886	0.998	5.741	
1,753.20	18.95	335.44	1,706.28	-157.32	-176.71	227.21	3.045	1.526	8.593	
1,792.90	20.66	344.61	1,743.64	-144.70	-181.25	219.06	2.676	1.292	6.929	
1,833.50	22.98	351.87	1,781.33	-129.94	-184.27	208.30	2.627	1.714	5.365	
1,874.00	25.39	357.35	1,818.28	-113.44	-185.79	195.25	2.438	1.785	4.059	
1,914.80	27.17	2.38	1,854.87	-95.39	-185.81	180.09	2.095	1.309	3.699	
1,955.10	29.95	3.22	1,890.26	-76.15	-184.86	163.40	2.091	2.069	0.625	
1,995.50	30.21	3.94	1,925.22	-55.94	-183.60	145.73	0.330	0.193	0.535	
2,035.60	30.21	4.37	1,959.87	-35.81	-182.14	128.02	0.162	0.000	0.322	
2,075.60	30.20	4.60	1,994.44	-15.75	-180.56	110.31	0.087	-0.007	0.172	
2,116.00	29.79	3.84	2,029.43	4.39	-179.07	92.57	0.415	-0.304	-0.564	
2,156.50	29.75	3.56	2,064.59	24.46	-177.78	75.00	0.107	-0.030	-0.207	

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Wellbore:	F-15D	Survey Calculation Method:	Minimum Curvature
Design:	F-15D	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,196.70	29.77	4.21	2,099.48	44.37	-176.43	57.53	0.241	0.015	0.485
2,277.40	29.78	4.76	2,169.53	84.32	-173.29	22.25	0.102	0.004	0.204
2,318.10	29.89	3.51	2,204.84	104.51	-171.83	4.49	0.465	0.081	-0.921
2,334.70	29.84	3.86	2,219.23	112.76	-171.30	-2.73	0.328	-0.090	0.633
2,398.90	29.97	9.05	2,274.89	144.54	-167.70	-31.39	1.210	0.061	2.425
2,438.60	29.98	13.28	2,309.29	163.99	-163.86	-49.82	1.597	0.008	3.196
2,477.50	29.97	17.83	2,342.99	182.70	-158.66	-68.37	1.753	-0.008	3.509
2,519.90	29.99	25.30	2,379.73	202.37	-150.88	-89.11	2.640	0.014	5.285
2,560.90	30.41	32.46	2,415.17	220.39	-140.93	-109.66	2.652	0.307	5.239
2,599.90	29.95	36.04	2,448.89	236.60	-129.90	-129.25	1.429	-0.354	2.754
2,630.60	30.28	37.52	2,475.45	248.93	-120.68	-144.61	0.794	0.322	1.446
2,670.50	34.29	42.15	2,509.18	265.25	-107.00	-165.74	3.540	3.015	3.481
2,711.40	38.53	44.94	2,542.09	282.82	-90.26	-189.57	3.338	3.110	2.046
2,752.10	42.00	46.98	2,573.14	301.09	-71.35	-215.18	2.736	2.558	1.504
2,792.40	46.02	47.94	2,602.12	320.01	-50.72	-242.26	3.033	2.993	0.715
2,832.70	50.32	47.55	2,628.99	340.20	-28.50	-271.27	3.208	3.201	-0.290
2,873.30	54.93	46.75	2,653.63	362.14	-4.86	-302.52	3.439	3.406	-0.591
2,913.30	59.10	46.48	2,675.40	385.18	19.52	-335.09	3.132	3.127	-0.202
2,953.60	62.83	48.54	2,694.96	408.97	45.50	-369.16	3.083	2.777	1.533
2,993.30	62.91	52.42	2,713.07	431.45	72.75	-402.82	2.610	0.060	2.932
3,033.80	62.98	56.67	2,731.50	452.36	102.12	-436.31	2.804	0.052	3.148
3,074.50	62.98	61.23	2,749.99	471.05	133.17	-468.84	2.994	0.000	3.361
3,114.70	63.50	66.91	2,768.11	486.74	165.44	-499.50	3.804	0.388	4.239
3,154.90	63.52	71.45	2,786.04	499.52	199.06	-528.46	3.032	0.015	3.388
3,195.20	63.59	75.71	2,804.00	509.72	233.66	-555.77	2.840	0.052	3.171
3,235.30	65.19	81.10	2,821.34	516.97	269.06	-581.05	3.828	1.197	4.032
3,275.10	66.93	85.66	2,837.50	521.15	305.18	-604.13	3.404	1.312	3.437
3,354.30	67.34	89.38	2,868.28	524.31	378.07	-646.27	1.308	0.155	1.409
3,395.50	68.96	92.97	2,883.62	523.52	416.30	-666.31	2.698	1.180	2.614
3,436.60	69.87	96.73	2,898.07	520.26	454.63	-684.34	2.654	0.664	2.745
3,477.60	71.37	101.82	2,911.69	514.02	492.78	-699.77	3.681	1.098	3.724
3,516.50	74.54	106.98	2,923.09	504.76	528.78	-711.49	4.522	2.445	3.979
3,556.80	73.24	107.38	2,934.28	493.33	565.77	-721.91	1.009	-0.968	0.298
3,597.70	76.60	107.41	2,944.92	481.52	603.45	-732.40	2.465	2.465	0.022
3,637.60	78.44	107.46	2,953.54	469.85	640.62	-742.73	1.384	1.383	0.038
3,678.50	79.55	108.18	2,961.35	457.56	678.84	-753.10	0.965	0.814	0.528
3,718.30	78.51	106.58	2,968.92	445.89	716.12	-763.49	1.420	-0.784	-1.206
3,758.80	73.62	104.51	2,978.67	435.35	753.98	-775.14	3.916	-3.622	-1.533
3,798.60	68.99	105.01	2,991.42	425.75	790.42	-786.82	3.508	-3.490	0.377
3,834.90	65.90	107.78	3,005.35	416.30	822.58	-796.29	3.315	-2.554	2.289
3,878.40	65.99	109.35	3,023.08	403.66	860.23	-806.06	0.991	0.062	1.083
3,918.70	65.98	109.47	3,039.48	391.42	894.95	-814.59	0.082	-0.007	0.089
3,959.10	68.18	111.60	3,055.22	378.37	929.79	-822.49	2.189	1.634	1.582
3,999.80	71.27	111.64	3,069.32	364.30	965.28	-829.89	2.278	2.278	0.029

Statoil Survey Report

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4,041.10	74.92	112.17	3,081.33	349.56	1,001.93	-837.35	2.677	2.651	0.385
4,081.40	74.89	111.29	3,091.82	335.16	1,038.08	-844.83	0.633	-0.022	-0.655
4,121.40	75.02	109.65	3,102.20	321.65	1,074.27	-853.08	1.192	0.097	-1.230
4,161.60	76.92	109.69	3,111.95	308.52	1,110.99	-861.94	1.418	1.418	0.030
4,174.20	77.15	109.31	3,114.78	304.42	1,122.56	-864.76	1.038	0.548	-0.905
4,202.30	78.85	108.76	3,120.62	295.46	1,148.55	-871.30	1.904	1.815	-0.587
4,242.10	79.01	107.19	3,128.26	283.40	1,185.70	-881.30	1.168	0.121	-1.183
4,282.70	78.90	106.62	3,136.04	271.82	1,223.82	-892.21	0.421	-0.081	-0.421
4,322.70	78.97	106.47	3,143.72	260.64	1,261.46	-903.20	0.122	0.052	-0.112
4,362.40	77.50	106.91	3,151.81	249.48	1,298.68	-913.99	1.158	-1.111	0.332
4,403.70	75.94	107.23	3,161.30	237.68	1,337.11	-924.89	1.156	-1.133	0.232
4,443.40	78.75	107.82	3,169.99	226.01	1,374.04	-935.09	2.168	2.123	0.446
4,483.80	79.04	108.06	3,177.78	213.80	1,411.76	-945.26	0.277	0.215	0.178
4,524.20	77.65	107.51	3,185.94	201.72	1,449.43	-955.51	1.107	-1.032	-0.408
4,564.90	80.05	107.84	3,193.81	189.59	1,487.47	-965.93	1.785	1.769	0.243
4,605.40	82.04	108.48	3,200.11	177.13	1,525.49	-976.04	1.547	1.474	0.474
4,645.70	82.04	108.37	3,205.69	164.51	1,563.35	-985.95	0.081	0.000	-0.082
4,671.00	79.48	107.27	3,209.75	156.87	1,587.12	-992.40	3.297	-3.036	-1.304
4,685.00	79.48	107.27	3,212.31	152.78	1,600.27	-996.09	0.000	0.000	0.000

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.60	220.60	30"	30.000	36.000	
1,368.40	1,340.36	20"	20.000	26.000	
2,612.00	2,459.37	13 3/8"	13.375	17.500	
3,294.70	2,845.16	10 3/4"x9 5/8"	9.625	12.250	
4,683.60	3,212.05	7"	7.000	8.500	

Checked By: _____ Approved By: _____ Date: _____